



9 March 2026

CERTIFICATE OF ANALYSIS

Customer: Northrop Grumman Systems Corporation

Purchase Order No: 5000679536 Part Number: MSRR4040-VAFR

Material: F3C 433/M65: 38%; 285AW; 50"

Specification: MSRR4040, Issue 4, dated November 2012.

Certificate of Conformance: Hexcel lot 92M0110713 has been manufactured and tested in conformance with specification MSRR4040, Issue 4, dated November 2012.

Lot No: 92M0110713 Date of Manufacture: 18 February 2026

Storage Life: 12 months from date of manufacture when stored at 0°.

Resin Lot No: 92M0111313 Manufactured by: Hexcel

Glass Lot No: 31W0027484 Manufactured by: Hexcel

- I. Prepreg Physical, Chemical and Mechanical Properties
See Certificate of Analysis Report.
- II. Individual Spool Physical Properties
See In Process Control Acceptance Testing Data.

Note: See Packing List for date of shipment, actual quantity and spools shipped.

Laney Defa 

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0110713**
Order Number: **92M0110713**

Report Printed: 25-Mar-2026 9:24:45AM
Status: **PASS**
Approval: **TUCKERRR**

1. Tensile, SBS, and TG Panel

ASTM D8336 Nottingham Tack

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713001/001	Tack Force (lbf)		1.878 - 1.962 - 2.731
	Tack Force Avg (lbf)		2.190

Dimensional Fiber Volume

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713001/001	Laminate Density By Immersion Avg (g/cc)		1.5086
	Laminate Density By Immersion Std (g/cc)		0.0020
	Laminate Density By Immersion CoV		0.13
	Dimensional Fiber Volume (%)		52.34 - 52.46 - 52.68
	Dimensional Fiber Volume Avg (%)	50 - 56	52
	Dimensional Fiber Volume Std (%)		0.172
	Dimensional Fiber Volume CoV		0.33
	Void Volume (%)		0.7 - 0.9 - 0.7
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.142
	Void Volume CoV		18.18
92M0110713012/002	Laminate Density By Immersion Avg (g/cc)		1.5089
	Laminate Density By Immersion Std (g/cc)		0.0015
	Laminate Density By Immersion CoV		0.10
	Dimensional Fiber Volume (%)		52.52 - 52.70 - 52.99
	Dimensional Fiber Volume Avg (%)	50 - 56	53
	Dimensional Fiber Volume Std (%)		0.237
	Dimensional Fiber Volume CoV		0.45
	Void Volume (%)		0.9 - 0.8 - 0.8
	Void Volume Avg (%)	MAX= 2	1
	Void Volume Std (%)		0.020
	Void Volume CoV		2.35

Short Beam Shear Stress-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713001/001	SBS Strength (MPa)		79.9 - 91.0 - 90.9 - 91.0 - 90.2

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0110713**
Order Number: **92M0110713**

Report Printed: 25-Mar-2026 9:24:45AM
Status: **PASS**
Approval: **TUCKERRR**

Short Beam Shear Stress-Warp-Ambient (Continued)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713001/001	SBS Strength Avg (MPa)	MIN= 65	89
	SBS Strength Std (MPa)		4.86
	SBS Strength CoV		5.49
92M0110713012/002	SBS Strength (MPa)		82.4 - 86.3 - 84.2 - 74.2 - 90.0
	SBS Strength Avg (MPa)	MIN= 65	83
	SBS Strength Std (MPa)		5.91
	SBS Strength CoV		7.08

Short Beam Shear Stress-Warp-150C(302F)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713001/001	SBS Strength (MPa)		60.4 - 57.3 - 59.2 - 61.2 - 62.8
	SBS Strength Avg (MPa)	MIN= 50	60
	SBS Strength Std (MPa)		2.08
	SBS Strength CoV		3.46
92M0110713012/002	SBS Strength (MPa)		64.9 - 64.1 - 66.6 - 64.1 - 68.1
	SBS Strength Avg (MPa)	MIN= 50	66
	SBS Strength Std (MPa)		1.73
	SBS Strength CoV		2.63

Tg By DMA

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713001/001	Tg by DMA in °C (° C)		266.1
	Tg by DMA in °C Avg (° C)	MIN= 225.0	266.1
92M0110713012/002	Tg by DMA in °C (° C)		259.8
	Tg by DMA in °C Avg (° C)	MIN= 225.0	259.8

Calculated Thickness/Ply

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713001/001	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30
	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		1.21
92M0110713012/002	Calculated Thickness/Ply Avg (mm)	0.28 - 0.32	0.30

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
 Configuration Number: **CN1301**
 Material Type: **F3C 433/M65;38%;285AW;50"**
 Lot Number: **92M0110713**
 Order Number: **92M0110713**

Report Printed: 25-Mar-2026 9:24:45AM
 Status: **PASS**
 Approval: **TUCKERRR**

Calculated Thickness/Ply (Continued)

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713012/002	Calculated Thickness/Ply StDev (mm)		0.0025
	Calculated Thickness/Ply CoV		1.09

Tensile Stress, Modulus, and Elongation-Warp-Ambient

Roll/Panel	Test/Result:	Requirements:	Values:
92M0110713001/001	Norm Tensile Strength (MPa)		980.0 - 885.8 - 881.2 - 958.8 - 932.5
	Norm Tensile Strength Avg (MPa)	MIN= 700	928
	Norm Tensile Strength Std (MPa)		43.71
	Norm Tensile Strength CoV		4.71
	Norm Modulus (GPa)		126.1 - 122.9 - 124.9 - 125.1 - 126.2
	Norm Modulus Avg (GPa)	MIN= 55	125
	Norm Modulus Std (GPa)		1.303
	Norm Modulus CoV		1.04
92M0110713012/002	Norm Tensile Strength (MPa)		932.3 - 917.7 - 988.3 - 956.6 - 991.0
	Norm Tensile Strength Avg (MPa)	MIN= 700	957
	Norm Tensile Strength Std (MPa)		32.74
	Norm Tensile Strength CoV		3.42
	Norm Modulus (GPa)		127.4 - 127.2 - 126.5 - 125.0 - 128.8
	Norm Modulus Avg (GPa)	MIN= 55	127
	Norm Modulus Std (GPa)		1.386
	Norm Modulus CoV		1.09

2. PHYSICAL PROPERTIES

Volatile content @100°C in pan

Roll	Test/Result:	Requirements:	Values:
92M0110713001	Volatile content @100°C in a Pan (%)		0.5 - 0.6
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	1
92M0110713012	Volatile content @100°C in a Pan (%)		0.6 - 0.6
	Volatile content @100°C in a Pan Avg (%)	MAX= 1	1

3. PHYSICAL TACK



Matrix Testing Laboratory
5400 S. 6700 W
West Valley City, Utah
84118

Certificate of Analysis Report

Specification: **MSRR4040 Rev. Issue 4**
Configuration Number: **CN1301**
Material Type: **F3C 433/M65;38%;285AW;50"**
Lot Number: **92M0110713**
Order Number: **92M0110713**

Report Printed: 25-Mar-2026 9:24:45AM

Status: **PASS**

Approval: **TUCKERRR**

Tack

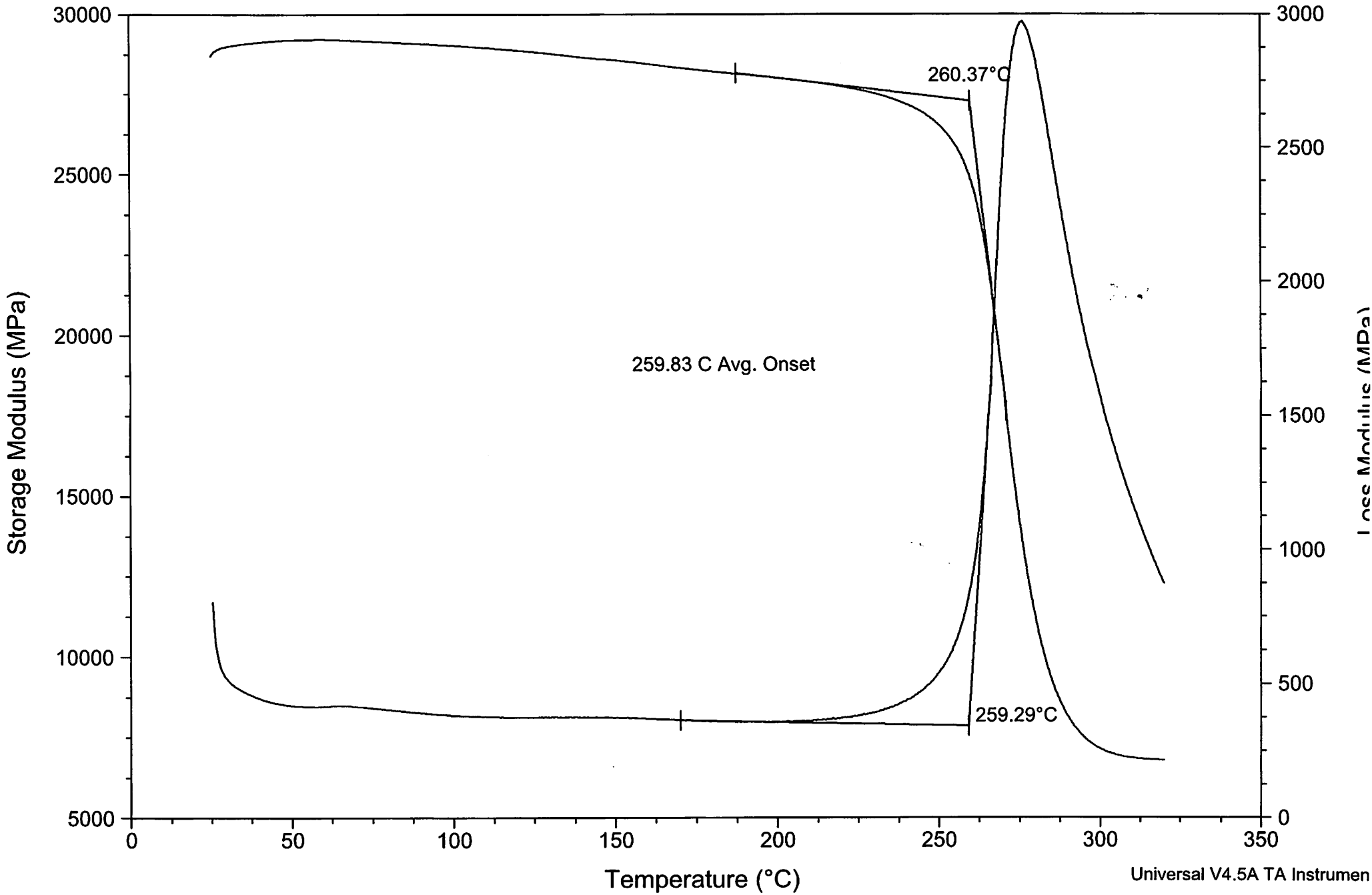
Roll	Test/Result:	Requirements:	Values:
92M0110713001	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0110713003	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0110713005	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0110713007	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0110713009	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30
92M0110713011	Tack Time to Failure (min.)	MIN= 30	30 - 30
	Tack Time to Failure Avg (min.)	MIN= 30	30

End of report

Sample: 92M0110713-002 SP#012
Size: 34.9200 x 9.9300 x 2.4300 mm
Method: Temperature Ramp

DMA

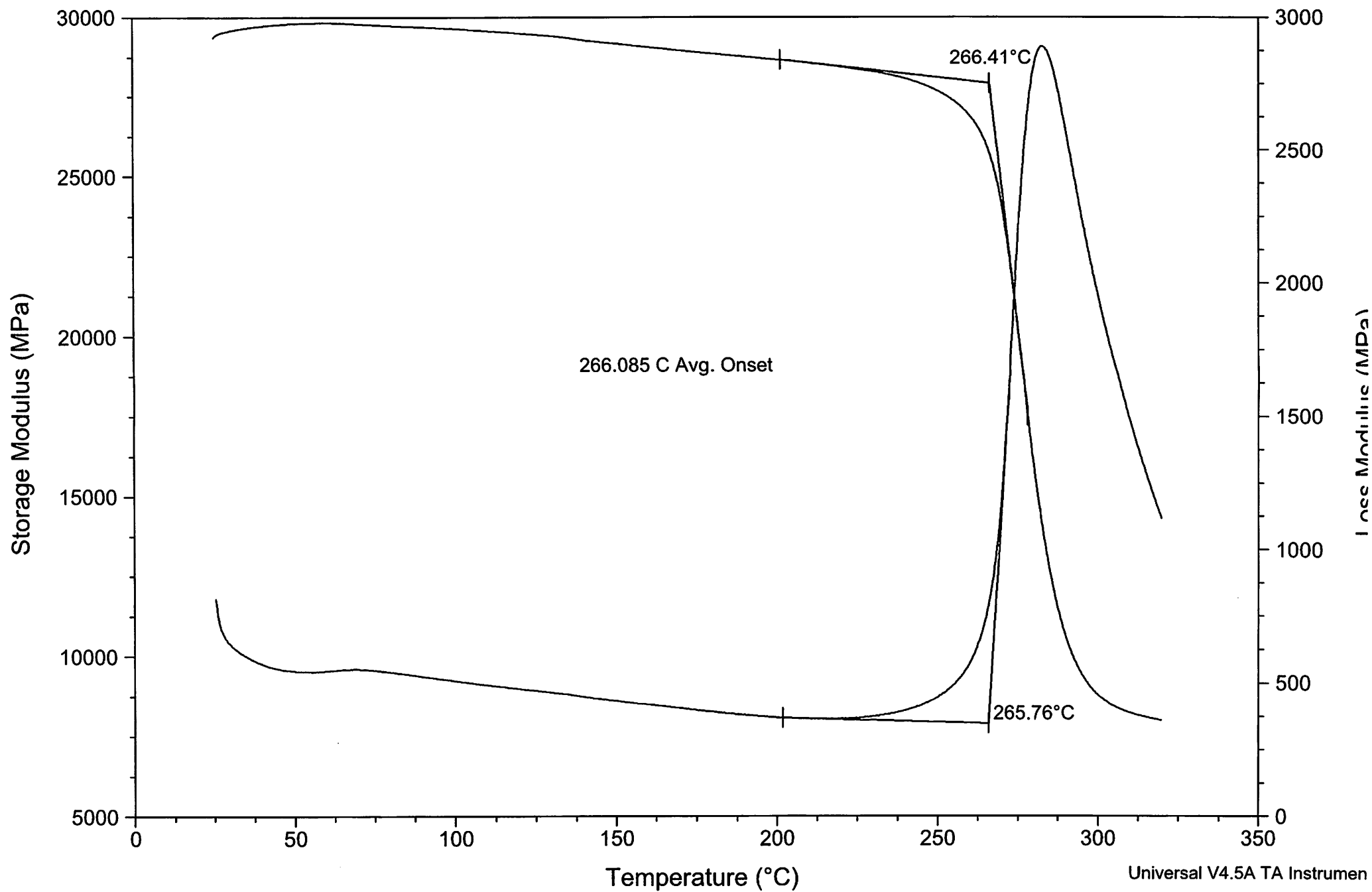
File: \\...\DMA Q800-2357\M65\92M0110713.002
Operator: KV
Run Date: 05-Mar-2026 08:00
Instrument: DMA Q800 V21.3 Build 96



Sample: 92M0110713-001 SP#001
Size: 34.9200 x 9.9300 x 2.4200 mm
Method: Temperature Ramp

DMA

File: \\...\DMA Q800-2357\M65\92M0110713.001
Operator: MC
Run Date: 05-Mar-2026 06:01
Instrument: DMA Q800 V21.3 Build 96



IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

RC / AWT Report for lot: **92M0110713**

3/5/2026

Page 1 of 3

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
92M 0110713001	0	1	4.5856	2.8030	38.9	280	458.6			SALYERBX	02/18/2026
		2	4.5928	2.7851	39.4	278	459.3			SALYERBX	02/18/2026
		3	4.5899	2.8082	38.8	281	459.0			SALYERBX	02/18/2026
		Avg.	4.5894		39	280	459	50.35	A	SALYERBX	02/18/2026
92M 0110713002	0	1	4.5904	2.8295	38.4	283	459.0			SALYERBX	02/18/2026
		2	4.6548	2.8836	38.1	288	465.5			SALYERBX	02/18/2026
		3	4.5997	2.8583	37.9	286	460.0			SALYERBX	02/18/2026
		Avg.	4.6150		38	286	462	50.30	A	SALYERBX	02/18/2026
92M 0110713003	0	1	4.5293	2.7835	38.5	278	452.9			HANSENAX	02/18/2026
		2	4.5186	2.8020	38.0	280	451.9			HANSENAX	02/18/2026
		3	4.5619	2.8206	38.2	282	456.2			HANSENAX	02/18/2026
		Avg.	4.5366		38	280	454	50.30	A	HANSENAX	02/18/2026
92M 0110713004	0	1	4.5616	2.8074	38.5	281	456.2			HANSENAX	02/18/2026
		2	4.5639	2.7982	38.7	280	456.4			HANSENAX	02/18/2026
		3	4.5104	2.7662	38.7	277	451.0			HANSENAX	02/18/2026
		Avg.	4.5453		39	279	454	50.29	A	HANSENAX	02/18/2026
92M 0110713005	0	1	4.5856	2.7860	39.2	279	458.6			HANSENAX	02/18/2026
		2	4.5704	2.7756	39.3	278	457.0			HANSENAX	02/18/2026
		3	4.6013	2.8008	39.1	280	460.1			HANSENAX	02/18/2026
		Avg.	4.5858		39	279	459	50.30	A	HANSENAX	02/18/2026
92M 0110713006	0	1	4.5620	2.7858	38.9	279	456.2			HANSENAX	02/18/2026
		2	4.5943	2.8101	38.8	281	459.4			HANSENAX	02/18/2026
		3	4.6126	2.8206	38.9	282	461.3			HANSENAX	02/18/2026
		Avg.	4.5896		39	281	459	50.25	A	HANSENAX	02/18/2026
92M 0110713007	0	1	4.5490	2.8226	38.0	282	454.9			HANSENAX	02/18/2026
		2	4.5043	2.8106	37.6	281	450.4			HANSENAX	02/18/2026
		3	4.6585	2.8986	37.8	290	465.8			HANSENAX	02/18/2026
		Avg.	4.5706		38	284	457	50.30	A	HANSENAX	02/18/2026
92M 0110713008	0	1	4.5356	2.8989	36.1	290	453.6			HANSENAX	02/18/2026
		2	4.5713	2.9543	35.4	295	457.1			HANSENAX	02/18/2026
		3	4.5491	2.9123	36.0	291	454.9			HANSENAX	02/18/2026
		Avg.	4.5520		36	292	455	50.30	A	HANSENAX	02/18/2026
92M 0110713009	0	1	4.5415	2.8686	36.8	287	454.2			HANSENAX	02/18/2026
		2	4.5830	2.8509	37.8	285	458.3			HANSENAX	02/18/2026
		3	4.5755	2.8681	37.3	287	457.6			HANSENAX	02/18/2026
		Avg.	4.5667		37	286	457	50.30	A	HANSENAX	02/18/2026
92M 0110713010	0	1	4.5072	2.8299	37.2	283	450.7			HANSENAX	02/18/2026
		2	4.5867	2.8599	37.6	286	458.7			HANSENAX	02/18/2026
		3	4.4383	2.7480	38.1	275	443.8			HANSENAX	02/18/2026
		Avg.	4.5107		38	281	451	50.30	A	HANSENAX	02/18/2026
92M 0110713011	0	1	4.4906	2.7794	38.1	278	449.1			HANSENAX	02/18/2026
		2	4.5834	2.8085	38.7	281	458.3			HANSENAX	02/18/2026
		3	4.4953	2.7775	38.2	278	449.5			HANSENAX	02/18/2026

ACCEPTANCE TESTING DATA

Page 2 of 3

Spool	Re - test	Rep	Init - Weight	Final Weight	RC	AWT	PAW	Width	Stat.	Tech	Date
=====											
Avg.			4.5231		38	279	452	50.30	A	HANSENAX	02/18/2026
=====											
92M	0	1	4.5617	2.8045	38.5	280	456.2			HANSENAX	02/18/2026
0110713012		2	4.6110	2.8070	39.1	281	461.1			HANSENAX	02/18/2026
		3	4.5656	2.8105	38.4	281	456.6			HANSENAX	02/18/2026
Avg.			4.5794		39	281	458	50.30	A	HANSENAX	02/18/2026
=====											

IN PROCESS CONTROL

ACCEPTANCE TESTING DATA

Lot Averages Report for Lot: **92M0110713**

03/05/2026

Page 3 of 3

Spool	Retest:	RC	AWT	PAW	Stamp
92M0110713001	0	39	280	459	A
92M0110713002	0	38	286	462	A
92M0110713003	0	38	280	454	A
92M0110713004	0	39	279	454	A
92M0110713005	0	39	279	459	A
92M0110713006	0	39	281	459	A
92M0110713007	0	38	284	457	A
92M0110713008	0	36	292	455	A
92M0110713009	0	37	286	457	A
92M0110713010	0	38	281	451	A
92M0110713011	0	38	279	452	A
92M0110713012	0	39	281	458	A
Averages:		38.1	282.3	456.0	



Product Certification

Page 1 of 2

COA Number:	SEG-COA-28801	Prod Order:	DOM:	Description:
Version:	1	31W0027484	18-Jul-2025	F3C433(FBL2)-50"
Issue Date:	09-Feb-2026	Weave Pattern:	5 Harness Satin	

Specifications

F3C433(FBL2) - GREIGE - 50inch - HRMS 111546 Rev. 6 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50inch - MSRR4040 Issue 4 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

F3C433(FBL2) - GREIGE - 50inch - MSRR9992 Issue 12 Warp Yarn HTA40 E13 3K 200tex Fill Yarn HTA40 E13 3K 200tex

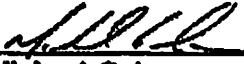
Rolls Tested

Fabric Testing Results

Serial#:	Prod Order:	Property:	Specification:	Result:
31G0000364652	31W0027484	Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000364653		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	282
31G0000364654		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.7
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	281
31G0000364655		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.4
		Dried FAW Avg	278 - 292 g/m ²	283
31G0000364656		Width	49.5 - 50.5 in	50.2
		Ends per inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per inch	17.0 - 18.5 Yarns Per Inch	17.4

This is to certify that material shipped against this order meets the specification(s) listed above.

This report should only be reproduced in its entirety
1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



Product Certification

Page 2 of 2

31G0000364656	31W0027484	Dried FAW Avg	278 - 292 g/m ²	281
31G0000364657		Width	49.5 - 50.5 In	50.2
		Ends per Inch	17.0 - 18.5 Yarns Per Inch	17.8
		Picks per Inch	17.0 - 18.5 Yarns Per Inch	17.5
		Dried FAW Avg	278 - 292 g/m ²	283

Certification Exceptions/Additional Information

QMS07045, REV. 1, SEC. 5E

WARP: 250120210

FILL: 250120210

This is to certify that material shipped against this order meets the specification(s) listed above.

This report should only be reproduced in its entirety
1913 N. King Street, Seguin, TX 78155
Telephone (830) 379-1580 Fax (830) 379-3814
www.hexcel.com


Michael Galvan
Certification Technician



TEIJIN CARBON AMERICA, INC.

121 Cardiff Valley Road
Rockwood, TN 37854
Tel. (865) 354-8408
Fax (865) 354-8409

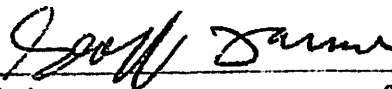
Certificate of Analysis

QAP 004 F2 Rev M

Customer:	Hexcel Reinforcement Corp (TX)	Contact:	Patrick Spieckermann; Amy Shehane; Jerry Tschoepe; Greg Early	Email/Fax:	Patrick.Spieckermann@hexcel.com; amy.shehane@hexcel.com; jerry.tschoepe@; Greg.Early@
CustomerPO#:	31PO0014263	Order:	SH001734	Ship Date:	7/1/2025
Customer P/N:	n/a	Product:	Tenax™-E HTA40 E13 3K 200tex		
Part Number:	G305303	Quantity:	10000.066 lb	Certifies To:	MS-X-CTC-105 Rev B; MSRR9317/11 Issue 1
Lot Number:	250120210				

We are pleased to advise that the product (s) listed above has been released for shipment. We certify that this lot of Carbon Fiber has been tested by standard test methods and determined to have the following properties:

	<u>Average</u>	<u>CV%</u>
Tensile Strength, MPA (ksi):	4145.8 (601)	4.2
Tensile Modulus, GPA (msi):	238.4 (34.6)	0.8
Elongation, %:	1.64	4.0
Linear Density (without size), g/m:	0.1994	1.0
Size, %:	1.27	7.3
Fiber Density, g/cc:	1.77	
Filament Diameter, µm:	7	
Twist, turn/m:	Never Twisted	
Date of Manufacture:	January 2025	
Manufacturing Location:	Oberbruch, Heinsberg, Germany	


Quality Assurance

8/30/2025

These commodities, technology or software are controlled by U. S. Export Administration Regulation. Diversion contrary to U. S. Law is prohibited.

If goods are not received in a reasonable time or if there are any questions, please call 1-800-252-3001 (or (865) 354-8408).
Teijin Carbon America, Inc. 121 Cardiff Valley Road, Rockwood, TN 37854.

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0110713	92M0110713001	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
-2	50	0	0.0	FUZZBALL
27	50	0	0.0	FUZZBALL
29	50	2	0.7	SNAG
73	50	0	0.0	FUZZBALL
92	50	0	0.0	FUZZBALL
101	50	0	0.0	FUZZBALL
117	50	0	0.0	FUZZBALL
119	50	0	0.0	FUZZBALL
125	50	0	0.0	FUZZBALL
170	50	0	0.0	FUZZBALL
192	50	24	8.3	CURED RESIN
201	50	1	0.3	CURED RESIN

Final Footage	Net Feet	Credit Method	Defect %	Date
215	212	A	1.047%	2/18/2026

Bag label - Hexcel Hexcel, 6700 W. 5400 S., SLC, Ut. 84116
F3C 433/M65;38%;285AW;50"

MSRR4040 Rev. ISSUE 4

Product: 111545 Batch: 92M0110713 Shelf life: 365 days DOM: 19-Feb-2026
QTY: 212.00ft Serial: 92M0110713001 Storage conditions: ≤ 0 F Expiration: 19-Feb-2027



Inspection Stamp



Date

2/18/2026

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713002	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
79	50	0	0.0	FUZZBALL
137	50	0	0.0	FUZZBALL
199	50	0	0.0	FUZZBALL

Final Footage 250	Net Feet 250	Credit Method A	Defect % 0.000%	Date 2/18/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/18/2026
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713003	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
118	50	0	0.0	FUZZBALL
126	50	0	0.0	FUZZBALL
135	50	60	20.8	CREASE
164	50	0	0.0	FUZZBALL
181	50	0	0.0	FUZZBALL
188	50	0	0.0	FUZZBALL
191	50	0	0.0	FUZZBALL
196	50	0	0.0	FUZZBALL

Final Footage 225	Net Feet 220	Credit Method A	Defect % 2.222%	Date 2/18/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/18/2026
--	--------------------------

ROLL CONDITION LOG

Batch Number	Serial Number	Fiber	Resin	Prepreg Width
92M0110713	92M0110713004	F3C 433	M65	50

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
18	50	0	0.0	FUZZBALL
94	50	216	75.0	CREASE

Final Footage	Net Feet	Credit Method	Defect %	Date
228	210	A	7.895%	2/18/2026

Inspection Stamp	Date
	2/18/2026

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713005	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
41	50	2	0.7	SLUB
51	50	0	0.0	FUZZBALL
123	50	0	0.0	FUZZBALL
174	50	0	0.0	FUZZBALL

Final Footage 225	Net Feet 224	Credit Method A	Defect % 0.074%	Date 2/18/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/18/2026
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713006	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
31	50	2	0.7	FUZZBALL
38	50	6	2.1	MISPICK
44	50	0	0.0	FUZZBALL
68	50	24	8.3	CREASE
74	50	2	0.7	FUZZBALL
199	50	0	0.0	FUZZBALL

Final Footage 224	Net Feet 221	Credit Method A	Defect % 1.265%	Date 2/18/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/18/2026
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713007	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
4	50	24	8.3	CURED RESIN
12	50	0	0.0	FUZZBALL
61	50	0	0.0	FUZZBALL
75	50	0	0.0	FUZZBALL
81	50	0	0.0	FUZZBALL
188	50	0	0.0	FUZZBALL

Final Footage 228	Net Feet 226	Credit Method A	Defect % 0.877%	Date 2/18/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/18/2026
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713008	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
80	50	0	0.0	FUZZBALL
159	50	24	8.3	CREASE

Final Footage 200	Net Feet 198	Credit Method A	Defect % 1.000%	Date 2/18/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/18/2026
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713009	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
15	50	0	0.0	FUZZBALL
26	50	1	0.3	MISSING PICK
152	50	0	0.0	FUZZBALL
191	50	2	0.7	FUZZBALL
200	50	6	2.1	RESIN LEAN

Final Footage 224	Net Feet 223	Credit Method A	Defect % 0.335%	Date 2/18/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/18/2026
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713010	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
153	50	0	0.0	FUZZBALL
166	50	0	0.0	FUZZBALL

Final Footage 222	Net Feet 222	Credit Method A	Defect % 0.000%	Date 2/18/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/18/2026
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713011	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
12	50	0	0.0	FUZZBALL
36	50	60	20.8	SNAG
63	50	2	0.7	FUZZBALL
92	50	0	0.0	FUZZBALL
112	50	2	0.7	FUZZBALL

Final Footage 208	Net Feet 202	Credit Method A	Defect % 2.564%	Date 2/19/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/19/2026
---	--------------------------

ROLL CONDITION LOG

Batch Number 92M0110713	Serial Number 92M0110713012	Fiber F3C 433	Resin M65	Prepreg Width 50
-----------------------------------	---------------------------------------	-------------------------	---------------------	----------------------------

Feet From Start	Defect Width (inches)	Defect Length (inches)	Credit Amount (sq. ft.)	Defect Type Remarks
47	50	0	0.0	FUZZBALL
59	50	0	0.0	FUZZBALL
98	50	0	0.0	FUZZBALL
141	50	2	0.7	FUZZBALL
164	50	0	0.0	FUZZBALL
171	50	0	0.0	FUZZBALL
198	50	0	0.0	FUZZBALL
230	50	6	2.1	FUZZBALL

Final Footage 234	Net Feet 233	Credit Method A	Defect % 0.285%	Date 2/19/2026
-----------------------------	------------------------	---------------------------	---------------------------	--------------------------

Inspection Stamp 	Date 2/19/2026
---	--------------------------